

An Introduction to Cassandra



A database is an organised collection of data that is generally stored and accessed electronically from a computer system. There are numerous databases that have been developed to meet different needs. This article discusses the various aspects of the Cassandra DB – its architecture, its components, read/write operations and a few of its query language syntaxes, with explanations.

Cassandra is a free and open source distributed database that has been designed and developed to handle a large amount of structured, semi-structured and unstructured data. It is an Apache product based on the wide-column NoSQL database, and it can re-collect/recapture and store data. It supports an easy replication process. Cassandra is written in Java, and is used for time-series data such as chats, logs and so on.

Here are some key reasons for using this database:

- Simple interface without any complex joins
- Non-relational and schema-less model
- High fulfilment and fast performance

Some basic information about Cassandra

Stable release	3.11.3/August 1, 2018
Written in	Java
Website	<i>Cassandra.apache.org</i>
Type	NoSQL database, data store
Main features	Distributed, scalable, a query language with fault tolerance. It supports replication across multiple data centres.

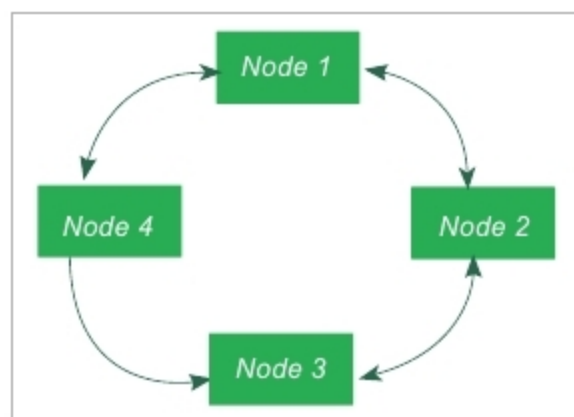


Figure 1: Cassandra ring architecture

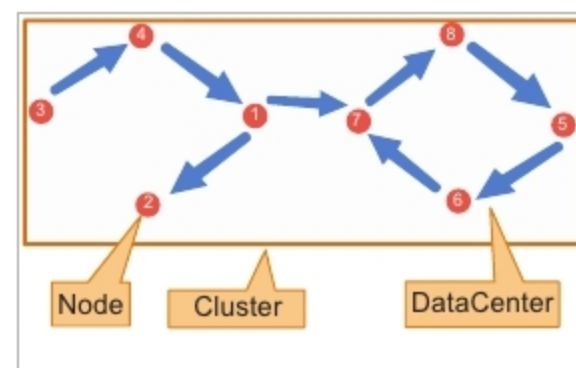


Figure 2: The relationship between components (the node, cluster and data centre)